DeepDetect: AI-Powered Credit Card Fraud Detection System

Abstract

Credit card fraud has become a major concern in the digital financial ecosystem, leading to significant financial losses for individuals and institutions. This project presents an AI-powered fraud detection system that identifies fraudulent transactions using machine learning techniques. The system analyzes transaction data and detects anomalies that may indicate fraud. By training models on historical transaction datasets, the system learns patterns associated with fraudulent behavior and provides real-time predictions. Such systems play a crucial role in improving financial security and minimizing fraud risks.

Introduction

With the increasing use of online payment systems, credit card fraud has grown rapidly, posing serious challenges to financial institutions. Fraudulent activities often involve complex patterns that are difficult to detect manually.

This project focuses on building an intelligent fraud detection system using machine learning algorithms. The system analyzes transaction data and classifies it as legitimate or fraudulent. One of the key challenges addressed in this project is handling imbalanced datasets, where fraudulent transactions represent only a small portion of the data. Advanced techniques and models are used to improve detection accuracy and reduce false positives.

Tools Used

The following tools and technologies were used:

- Programming Language: Python
- Libraries:
    - Pandas, NumPy (data processing)
    - Scikit-learn (machine learning models)
    - Matplotlib (data visualization)
- Techniques:
    - Classification Algorithms (e.g., Logistic Regression, Random Forest)

    - Data Preprocessing & Feature Scaling
    - SMOTE (handling imbalanced data)
- Platform: Jupyter Notebook / Streamlit

Steps Involved

1. Data Collection
    - Collected credit card transaction dataset
2. Data Preprocessing
    - Cleaned data and handled missing values
    - Performed feature scaling and normalization
3. Handling Imbalanced Data
    - Applied techniques like SMOTE to balance the dataset
4. Feature Engineering
    - Selected relevant features for model training
5. Model Development
    - Trained machine learning models to classify transactions
6. Model Evaluation
    - Evaluated using metrics like accuracy, precision, recall
7. Deployment
    - Integrated model into a Streamlit interface for real-time prediction

Conclusion

The DeepDetect system successfully demonstrates the application of machine learning in detecting credit card fraud. By analyzing transaction patterns and identifying anomalies, the system enhances financial security and reduces fraudulent activities. While the current model achieves good accuracy, future improvements can include deep learning models, real-time streaming data processing, and integration with banking systems. Overall, this project highlights the importance of AI in safeguarding digital financial transactions.